

the shared memory was in use by the low priority meteorological data gathering task. Again and again before it can release this shared memory it is interrupted because the high priority information management task needed to be executed and it has the next time slot scheduled to it. But, the information management task now needed access to shared memory which is now being held by the low priority meteorological data gathering task. So as mentioned before, the information management waits politely for the shared memory to be released. Now, the meteorological data gathering has to run again. However, before it can finally release the shared memory the medium priority data transmission task wants to join the party and is scheduled. This task then takes extremely long to finish because of all these interruptions. But now the low priority task just cannot release the shared memory because it will not get the chance to be scheduled because of the higher priority tasks which are waiting. Now as a result both the high priority and the low priority tasks are locked and are incapable of doing anything. The system sees that a high priority task is waiting for too long and panics and resets the system, resulting in loss of all that collected data, and numerous groans from NASA on Earth waiting for the results of this expensive expedition.

Ariane 5

Ariane 5, Flight 501, was launched on June 4, 1996, and has the dubious honor of being the first unsuccessful European test flight and of also involving one of the most infamous computer bugs in history. There was a malfunction in the control software, because of which the rocket suddenly went off its assigned flight path 37 seconds after the launch and then subsequently committed hara-kiri due to its automated self-destruct system when the core of the vehicle disintegrated due to high aerodynamic forces. Phew.

An operand error had occurred. This happened due to a very unexpected high value of internal alignment function result call the BH, or the Horizontal Bias. (yes we know that it should be HB but try explaining that to the people who created it). The Horizontal Bias is related to the horizontal velocity which is sensed by the platform, and this value is calculated as an indicator, over time, for alignment accuracy. What happened here was that the value of BH was much higher than what it was expected to be because the initial stage of the trajectory of Ariane 5 was different from that of the previous Ariane, Ariane 4, and this resulted in much higher horizontal velocity values. During execution of a data conversion from 64-bit floating point to